Luba Ira Korlat

Software Engineer | Scientific Computing | Full-Stack and Flutter Development

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Professional Summary

Software engineer with an M.Sc. and experience in scientific software, applications and teaching. Develops Python, web and Flutter solutions alongside manufacturing process support.

Technical Skills

Languages: Python, TypeScript, Dart, Java, C, C++, C#, SQL, R, HTML, CSS
Web and mobile: Flutter, React, Next.js, Phaser, FastAPI, Django, Node.js, REST APIs
Scientific computing and AI: NumPy, SciPy, TensorFlow, PyTorch, PINNs, NLP, OpenCV
Data: PostgreSQL, MySQL, MongoDB, Firebase, Supabase
Tools: Git, Docker, Linux, CI/CD, pytest, Playwright, Vitest

Professional Experience

Academic Transcriber and Tutor | Aleh Foundation

October 2026 - Present | Previous tenure: September 2023 - December 2025

Transcribe academic lectures, prepare accessible material and provide individual tutoring.

Process Support - AOI Program Configuration | Flex

June 2026 - Present

Configure automated optical inspection (AOI) programs and support production processes.

Software Developer | Self-Employed

January 2025 - Present

Develop mobile apps, websites, backend services and technical prototypes for private clients.

Computer Science and Mathematics Tutor | ALUT and Ono Academic College

January 2026 - June 2026

Tutored students on the autism spectrum through structured explanations and practical exercises.

Instructor | Cyber Education Centre - Magshimim Program

November 2025 - June 2026

Taught networks, sockets and client-server programming; reviewed and debugged student code.

Software Engineering Tutor | Sami Shamoon College of Engineering

April 2023 - May 2024

Tutored algorithms, data structures, object-oriented programming and software engineering.

Database Practice Tutor | Achva Academic College

November 2021 - April 2022

Led practice sessions in SQL, relational modeling and database normalization.

Education and Research

M.Sc. Software Engineering | Sami Shamoon College of Engineering

2022 - 2024 | Graduated with honors | GPA: 96/100

Thesis: Two-phase Stefan problem for lunar regolith melting.
Research: numerical heat transfer, phase change and physics-informed neural networks.

B.Sc. Software Engineering | Sami Shamoon College of Engineering

Graduated 2022 | GPA: 80/100

Final project: Hebrew text summarization using natural language processing.

National Service

Administrative Secretary | Beersheba Magistrate Court | September 2011 - September 2012

Languages: Hebrew, English and Russian - fluent

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Selected Projects

Independent scientific, space, application and game-development projects.

Heat Equation and Magnetic Torus Simulators

Python | NumPy | SciPy | TensorFlow | PyTorch | PyQt6 | PyVista
Developed a heat-equation and two-phase Stefan solver with physics-informed neural networks.
Added experimental torus thermal models and interactive 3D visualization.
github.com/lubacareer/HeatEquationSimulator

Europa Engine

TypeScript | React | Electron | Three.js | WebGPU
Built an experimental Jupiter-and-three-moon simulation using JPL Horizons initial states.
Added 3D visualization, conservation diagnostics and separate ice-ocean thermal experiments.
github.com/lubacareer/jupiter

NISAR Surface Change

TypeScript | React | Python | Electron | ECharts
Built a NISAR-inspired prototype using reproducible synthetic radar data.
Linked time-series charts and before/after comparisons through validated data contracts.
github.com/lubacareer/Nasa2

SMD Twin Lab

Python | PySide6 | NumPy | PyQtGraph
Developed an electronics learning prototype with circuit simulation and component-fault injection.
Added waveform visualization and structured reports for a reference sensor board.
github.com/lubacareer/electricity

Financial Diary

Dart | Flutter | Tesseract OCR | AES-GCM
Built an Android and Windows finance app with encrypted storage and offline statement OCR.
Implemented transaction reconciliation, cash-flow estimates and English, Hebrew and Russian UI.
github.com/lubacareer/finance1

The Case of the Missing Logic

TypeScript | Phaser | Vite | Vitest | Playwright
Built a playable point-and-click mystery prototype with dialogue, inventory and puzzle progression.
Implemented polygon-based navigation and automated puzzle-flow tests.
github.com/lubacareer/CML

Spritebury

TypeScript | React | Next.js | Supabase | PostgreSQL
Developed a virtual-city game prototype with accounts, avatars, profiles and inventory.
Implemented server-validated marketplace purchases and database access policies.
github.com/lubacareer/spritebury

ccomp - C Subset Compiler

C | Flex | Bison | Make
Implemented parsing, syntax trees, semantic checks and three-address code generation.
Added a basic x86-64 assembly emitter for an educational C-subset compiler.
github.com/lubacareer/ccomp

Trend Detective - Independent NASA Data Project

In development | Public distribution overview
Prepared the public distribution repository for a desktop NASA-observation project.
github.com/lubacareer/nasaCLPS-releases

My Portfolio Website: portfolio-blush-five-82.vercel.app